

price, volume, or depth-based criteria) are not included.²⁸⁵ Accordingly, the Commission is not adopting factors other than the TWAQS to measure which NMS stocks would be assigned a minimum pricing increment of \$0.005.

One commenter suggested that issuers should be able to select the minimum pricing increment for the quotes and orders of their stock.²⁸⁶ The Commission disagrees. Rule 612 is an important rule under Regulation NMS and serves to link the markets within the national market system by establishing uniform minimum pricing increments for all NMS stocks. This important linkage function would be undermined by allowing increments to be individually assigned to each NMS stock in a non-uniform manner. Rule 612, as originally adopted and as amended, standardizes minimum pricing increments based on transparent and objective criteria in order to ensure that minimum pricing increments are applied uniformly. Introducing issuer choice would eliminate such standardization and enable individual issuers to choose different minimum pricing increments based on their specific, unique individual preferences which would likely result in random and inconsistent application of increments across NMS stocks that otherwise share several relevant trading characteristics. Minimum pricing increment for quotes and orders of NMS stocks priced greater than, or equal to, \$1.00 per share based on unpredictable, opaque, non-standard criteria of individual issuers would result in unnecessary complication, such as varied minimum quoting increments and investor confusion, to the national market system. Further, market participants would likely incur additional costs related to, for example, the monitoring and tracking of the minimum pricing increments for issuers.

²⁸⁵ See infra section VII.D.1.b.iii.

²⁸⁶ See Angel Letter at 5. But see Harris Letter at 8 (opposing suggestions that issuers should choose ticks).

7. Rule 612(a) - Definitions

As adopted, amended Rule 612(a) contains two definitions for purposes of the rule – “Evaluation Period” and “Time Weighted Average Quoted Spread.” The primary listing exchanges will use these definitions in identifying the required minimum pricing increments for NMS stocks.

a. Evaluation Period

The Commission proposed to define “Evaluation Period” as the last month of a calendar quarter (March in the first quarter, June in the second quarter, September in the third quarter and December in the fourth quarter) of a calendar year during which the primary listing exchange shall measure the TWAQS of an NMS stock that is priced equal to, or greater than, \$1.00 to determine the minimum pricing increment to be in effect for the next calendar quarter, as set forth by proposed paragraph (c). In other words, the minimum pricing increment for quotes and orders would have been evaluated every quarter based on one month’s worth of data and could have potentially changed once every quarter.

After considering the comments, the Commission is adopting a revised definition of Evaluation Period. Rule 612(a)(1) defines Evaluation Period as (i) the three months from January through March of a calendar year and (ii) the three months from July through September of a calendar year during which the TWAQS of an NMS stock shall be measured by the primary listing exchange to determine the minimum pricing increment for each NMS stock.

The Commission received comments on the proposed definition of the Evaluation Period.²⁸⁷ Two commenters generally supported the definition as proposed.²⁸⁸

Several commenters stated that one month was too short a period for measuring and calculating TWAQS.²⁸⁹ One commenter stated that an analysis of one month's data "has the potential to disproportionately weigh systemic and idiosyncratic events (including corporate actions) resulting in unrepresentative tick sizes."²⁹⁰ Another commenter stated that "longer evaluation periods will reduce the risk that short-term aberrations will have an outsized impact on market structure. Using too short of an evaluation period, especially during periods of heightened volatility, could lead to unrepresentative price variations that ultimately result in illogical minimum price increments."²⁹¹ A few commenters recommended providing a longer period for conducting data analysis. Specifically, one commenter suggested that the time frame be "coterminous with the time between tick size changes. In other words, if tick sizes are adjusted every quarter, then the evaluation period should be every quarter . . ."²⁹² Another

²⁸⁷ See, e.g., IEX Letter I; Optiver Letter; NYSE Letter I; Pragma Letter; MMI Letter; FIA PTG Letter II; BlackRock Letter; SIFMA Letter II; T. Rowe Price Letter; Cboe Letter I and Cboe Letter II; UBS Letter; and JPMorgan Letter at 4.

²⁸⁸ See, e.g., IEX Letter I and NYSE Letter I (stating that the quarterly updates based on the last month of a quarter's data would "ensure that the next quarter's universe of tick-constrained names is selected using the most recent and relevant basis and allows for monitoring of other securities that are not yet tick-constrained but maybe starting to exhibit tick-constrained behavior."). IEX, however, recommended that the second month of a quarter be used for calculating the TWAQS. See also *infra* note 305 and accompanying text. See also Harris Letter at 7.

²⁸⁹ See, e.g., Optiver Letter at 2; MMI Letter at 6; FIA PTG Letter II at 2; and UBS Letter at 13. See also Cboe Letter I at 5 (stating that the framework for reevaluating the parameters for revising tick changes according to their proposed methodology should be quarterly or bi-annually so that the parameters "remain nimble to changing market conditions.").

²⁹⁰ See Optiver Letter at 2.

²⁹¹ See FIA PTG Letter II at 2-3. See also MMI Letter at 6 ("The evaluation period preceding the change should be at least one quarter to avoid capturing instances of market volatility, or events such as stock splits that may indirectly drive trading interest that cause the behavior and characteristics of a stock to depart dramatically from its history.").

²⁹² See FIA PTG Letter II at 2-3.

commenter suggested that the Evaluation Period should be at least one quarter.²⁹³ Another commenter recommended that the Evaluation Period be performed on an annual basis so as to reduce costs and operational risks that may be created by needing to update relevant systems.²⁹⁴ Finally, one commenter suggested that the evaluation of NMS stocks be conducted on a semi-annual basis, based on six-months of data, to reduce variability and complexity.²⁹⁵

A few commenters provided suggestions as to the length of time between tick adjustments.²⁹⁶ One commenter stated that ticks should be adjusted on a monthly basis rather than a quarterly basis.²⁹⁷ The commenter stated “[w]e would expect more frequent smaller updates to reduce how often and how long a stock’s tick stays outside the optimal range.”²⁹⁸ Another commenter, however, suggested that the Commission align tick adjustments with other elements in the proposal. Specifically, this commenter recommended that “the Commission reduce the frequency of changes and synchronize the intervals for revising market structure parameters by updating both round lots and tick sizes on a quarterly or semi-annual basis.”²⁹⁹ Another commenter suggested an annual consideration as a means to reduce burdens on market participants and reduce operational risks.³⁰⁰

²⁹³ See MMI Letter at 6.

²⁹⁴ See UBS Letter at 13.

²⁹⁵ See JPMorgan Letter at 4.

²⁹⁶ See Pragma Letter; UBS Letter; and BlackRock Letter. See also SIFMA Letter II.

²⁹⁷ See Pragma Letter at 1.

²⁹⁸ See id.

²⁹⁹ See BlackRock Letter at 10. See also SIFMA Letter II (commenting on three different elements – ticks, access fee caps and round lots – that would have to be updated and stating “[b]roker-dealers will be required under the Tick Size Proposal to update their systems to appropriately account for all three of these variable changes, which carries inherent risks (and costs) of inadvertent errors relative to today’s environment where each of these variables are static.”). See also section V.B.3.b.iii for a discussion of the modifications to the round lot definition.

³⁰⁰ See UBS Letter at 12.

After considering the comments on the length of the Evaluation Period, the amended rule will require that the TWAQS be measured over a longer period of time than proposed, i.e., using three months' worth of trading data instead of one month, and minimum pricing increments will be assigned on a less frequent basis, i.e., every six months instead of every three months. The Commission conducted analysis to evaluate the length of the data analysis for the TWAQS and the length of time between minimum pricing increment assignments.³⁰¹ This revised definition balances the concerns raised by commenters that a TWAQS measured over too short a time frame could potentially be skewed by high volatility or unique events, such as corporate actions,³⁰² but that a TWAQs measured over too long of a time period would increase the probability of assigning a stale minimum pricing increment for quotes and orders that does not reflect the prevailing trading characteristics. Further, an annual evaluation would potentially cause some NMS stocks to remain in a sub-optimal minimum pricing increment for too long, while a monthly evaluation of NMS stocks would raise concerns about investor confusion with frequent re-assignments and increase operational risks due to the need for frequent systems updates. The adopted semiannual evaluation addresses the potential burdens and concerns of an Evaluation Period that is either too long or too short.

Finally, two commenters recommended adding an implementation time period between the calculation of the TWAQS and the potential change to an NMS stock's minimum pricing increment.³⁰³ One commenter stated that the proposed rule "leaves little to no time for the

³⁰¹ See infra section VII.D.1.d for additional discussion on the three-month period.

³⁰² See supra note 290 and accompanying text. The suggestion that minimum pricing increments be updated on a monthly basis would raise these concerns.

³⁰³ See T. Rowe Price Letter and IEX Letter.

industry to communicate the change and update systems to reflect the new tick sizes.”³⁰⁴ Both commenters suggested that the rule should provide one month between the end of the data collection and the effectiveness of any new minimum pricing increments.³⁰⁵ One commenter stated that one month between the calculation of the TWAQS and the implementation of new tick sizes would “give the industry adequate time to process changes and minimize errors.”³⁰⁶ Another commenter stated that the quarterly changes with short transition time would raise operational risk in the market and at individual firms.³⁰⁷ This commenter also stated that “[f]requent changes to tick sizes will require considerable investor education.”³⁰⁸ The Commission agrees with commenters’ suggestions and is also adopting an implementation period for introducing new minimum pricing increments after an Evaluation Period. As adopted, market participants will have one month to implement any new minimum pricing increments. This will reduce concerns about operational risk and will provide market participants with time to inform investors of any changes in placing orders. One month is an adequate time period for market participants to adapt and make any required systems change to reflect the change, if any, in minimum pricing increment of the quotes and orders of an NMS stock that is priced greater than, or equal to, \$1.00 per share. A longer period could partially nullify the objectives of the adopted rule to ameliorate issues related to the constraint of stocks that are quoting at the \$0.01 minimum pricing increment.

³⁰⁴ See T. Rowe Price Letter at 4.

³⁰⁵ See IEX Letter I at 7 (suggesting that the Commission provide “one month between the end of data collection and the beginning of trading with the reallocated tick sizes, in order to avoid any unanticipated disruptions.”) and T. Rowe Price Letter.

³⁰⁶ See T. Rowe Price Letter at 4.

³⁰⁷ See Fidelity Letter at 13.

³⁰⁸ See Fidelity Letter at 13. See also Harris Letter at 7 (recommending mechanisms to ensure traders can determine relevant ticks).

As discussed below, the Commission has aligned the semiannual evaluation and implementation of minimum pricing increments and the dates for implementing any minimum pricing increments with the timing for changes to NMS stocks round lot assignment.³⁰⁹ The Commission agrees with commenters who recommended that these two evaluations and updates be conducted at the same time. This will lessen the burdens on the primary listing exchanges and market participants of implementing new minimum pricing increments and round lots. Further, it will lessen operational risks associated with frequent system updates.

Rule 612(a)(1) also requires that the TWAQS be measured for all NMS stocks by the primary listing exchange. One commenter requested clarification as to what would occur in volatile situations “where a low-priced stock (e.g., sub \$1.00) suddenly jumps to a higher price (e.g., \$8.00).”³¹⁰ This situation could occur under preexisting Rule 612 as the relevant minimum pricing increment is based on the price of the order or quote. However, because orders in an NMS stock can be submitted with prices at or above \$1.00 and below \$1.00 depending on its current market price, under the amended rule, each NMS stock must have its TWAQS measured so that a minimum pricing increment will be assigned for those orders that are priced at or above \$1.00. Therefore, as amended, all NMS stocks will be assigned a minimum pricing increment based on its TWAQS and investors will be able to understand the relevant minimum pricing increment for their orders when priced at or over \$1.00 and when priced under \$1.00. Under the rule, as adopted, quotes and orders in NMS stocks that are priced less than \$1.00 will continue to have a minimum pricing increment of \$0.0001.³¹¹ The operation of the amended rule is

³⁰⁹ See also infra section V.B.3.b.iv.

³¹⁰ See SIFMA Letter I at 43.

³¹¹ See Rule 612(b)(3).

consistent with how the preexisting rule operates in that quotes and orders for a particular NMS stock may be required to be priced in a \$0.01 or \$0.005 increment when the price of an order is equal to or greater than \$1.00 and may also be priced in a \$0.0001 increment when the price of an order is less than \$1.00.

b. Time Weighted Average Quoted Spread

The Commission proposed to define TWAQS as the average dollar value difference between the NBB and NBO during regular trading hours where each instance of a unique NBB and a unique NBO is weighted by the length of time that the quote prevailed as the NBB or NBO. The Commission did not receive any comments on the definition of TWAQS.³¹² The definition in Rule 612(a)(2) is adopted as proposed.

8. Rule 612(b)(1) – Semiannual Operative Dates

Rule 612, as adopted, contains amended paragraph (b)(1), which defines the operative dates for the minimum pricing increments assigned to each NMS stock and provides a month-long time period to implement potentially new minimum pricing increments at the end of each Evaluation Period. Specifically, minimum pricing increments for quotes and orders will be operative on the first business day of May following the Evaluation Period from January through March and the first business day of November following the Evaluation Period from July through September.³¹³ In adopting these operative dates, the Commission seeks to reduce the risk that market participants may not be fully staffed during the time that technology changes are necessary to implement new minimum pricing increments. Further, in addition to providing

³¹² As discussed above, the Commission received comment on whether there should be factors in addition to the TWAQS for determining whether an NMS stock is tick-constrained. See supra section III.C.6.

³¹³ See supra section III.C.7.a, for a discussion of the adopted Evaluation Period.

market participants with adequate time to make necessary systems changes, the implementation period will also provide adequate time for investors to be notified about the minimum pricing increment for the quotes and orders of NMS stocks that are priced equal to or greater than \$1.00.³¹⁴

Two commenters requested clarification as to how stock splits would be handled.³¹⁵ Once assigned under Rule 612(b)(1), minimum pricing increments will remain operative until the next operative date (i.e., May or November). Therefore, a stock split will not impact an NMS stock's minimum pricing increment until the next cycle. In order to avoid the complexity and confusion that could occur if the minimum pricing increment of NMS stocks were reassigned at unpredictable times, minimum pricing increments will not be changed during the time between operative dates.

9. Rule 612(c) - New NMS Stocks

Commenters asked for clarification on how new NMS stocks would be handled under Rule 612.³¹⁶ One commenter questioned how new NMS stocks and IPOs would be assigned minimum pricing increments and stated that allowing exchanges to assign different initial tick sizes could lead to “arbitrage of issuers choosing the exchange that offers the most favorable initial tick size.”³¹⁷ Another commenter stated that “the simplest and most intuitive alternative would be to use specifications which were previously considered to be the standard unit, such as a \$0.01 tick size.”³¹⁸ The Commission agrees. New NMS stocks will be assigned the same initial

³¹⁴ See supra note 303 and accompanying text.

³¹⁵ See SIFMA Letter II at 42 and Virtu Letter II at 20.

³¹⁶ See SIFMA Letter II at 42; BlackRock Letter at 10; Virtu Letter II at 18.

³¹⁷ See SIFMA Letter II at 42; Virtu Letter II at 18.

³¹⁸ See BlackRock Letter at 10.

minimum pricing increment under Rule 612(c), which requires all securities that become an NMS stock to be assigned to the minimum pricing increment of \$0.01.³¹⁹ Thereafter, the TWAQS of the NMS stock will be calculated during the next Evaluation Period to determine which minimum pricing increment will be required under Rule 612(b)(2). Sub-penny increments are limited to quotes and orders priced \$1.00 or more for those NMS stocks that have a demonstrated narrow TWAQS during the defined Evaluation Period; new NMS stocks that become eligible for trading during an operative period will not satisfy this requirement. New NMS stocks will have their TWAQS calculated during the next Evaluation Period after they start trading.

10. Rule 600(b)(89) - Regulatory Data

The Commission proposed to amend the definition of regulatory data in Rule 600(b)(78)³²⁰ to require the primary listing exchange for each NMS stock to calculate and provide to competing consolidators, self-aggregators, and the exclusive SIPs an indicator of the applicable minimum pricing increment required under Rule 612. The Commission is adopting the minimum pricing increment indicator, as proposed, under the definition of regulatory data in Rule 600(b)(89)(i)(F). A minimum pricing increment indicator will be useful to market participants, including investors, by providing important information about the relevant minimum pricing increment for each NMS stock. This indicator will help market participants, including investors, with submitting orders in the relevant increment. Because the minimum pricing increment can change on a semiannual basis depending on the TWAQS on an NMS

³¹⁹ See also section V.B.3.b.iii.

³²⁰ Preexisting Rule 600(b)(78) was subsequently renumbered to Rule 600(b)(89) by the Rule 605 Amendments. See Rule 605 Amendments, supra note 10.